



AI INNOVATION HACKATHON 2026

Research & Publications

STANDARDIZED PROBLEM STATEMENTS

Women in AI, Research, Innovation & Entrepreneurship Club

Research • Innovation • Entrepreneurship

5 Challenges • Online • AI Innovation Hackathon 2026

Each challenge follows the same structured format for clarity, consistency and ease of review.

CHALLENGE PORTFOLIO

NO.	CHALLENGE	DIFFICULTY
1	AI-Assisted Literature Review & Research Gap Finder	Intermediate
2	AI Research Paper Quality & Citation Assistant	Intermediate to Advanced
3	AI Research Topic & Problem Statement Generator	Intermediate
4	AI Research Paper Summarizer & Key Findings Extractor	Intermediate
5	AI Research Collaboration & Paper Matching Platform	Intermediate to Advanced



CHALLENGE 01

AI-Assisted Literature Review & Research Gap Finder

DOMAIN

Research Discovery / Literature Review

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Students and early-stage researchers often struggle to efficiently review a large number of research papers, identify the most relevant studies, compare findings, and determine meaningful research gaps. Manual literature screening is time-consuming and important studies or connections may be missed.

EXPECTED SOLUTION

Develop an AI-powered tool that accepts a research topic or keywords, retrieves relevant research papers, summarizes key findings, organizes papers by themes, identifies relationships between studies, and suggests potential research gaps or unanswered questions.

SUGGESTED DATASETS / RESOURCES

OpenAlex; Semantic Scholar; Crossref; PubMed for biomedical topics; publicly available research papers.

AI / ML COMPONENT

NLP; semantic search; text embeddings; document clustering/topic modeling; summarization; Retrieval-Augmented Generation (RAG).

WHAT MAKES THE CHALLENGE INTERESTING?

Addresses a real research workflow problem and combines information retrieval, NLP, summarization and research-gap discovery.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter topic/keywords retrieve papers theme clusters structured summaries relationship view candidate gaps with linked sources.

TECHNICAL REQUIREMENTS

Python; scholarly-data/API handling; document retrieval/parsing; embeddings; clustering; summarization; source citation/traceability; dashboard.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Retrieve relevant papers and provide source references.
- Generate structured summaries and theme/grouping.
- Present candidate gaps or unanswered questions with evidence links.

OPTIONAL ADVANCED FEATURES

- Add citation-network visualization.
- Compare retrieval strategies.
- Add user feedback to refine results.
- Add claim/evidence traceability for generated gap suggestions.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use lawful/public or explicitly permitted research resources. Respect copyright, database terms and API limits. Clearly distinguish generated suggestions from verified research facts, preserve source attribution, and avoid fabricating citations, authors, findings or research gaps.

INNOVATION ANGLE

Connects literature discovery, synthesis and gap identification in one research workflow.

SUCCESS / EVALUATION METRICS

Retrieval relevance/precision, summary quality, source traceability, clustering coherence and usefulness of gap suggestions.



CHALLENGE 02

AI Research Paper Quality & Citation Assistant

DOMAIN

Academic Writing / Citation Support

DIFFICULTY

Intermediate to Advanced

PROBLEM STATEMENT

Researchers, especially beginners, often face difficulties with academic structure, clarity, logical flow, unsupported claims, and appropriate citation placement. It can also be difficult to determine whether references actually support the claims made in a research paper.

EXPECTED SOLUTION

Build an AI assistant that analyzes a research draft, provides feedback on academic writing and structure, detects potentially unsupported claims, identifies places where citations may be needed, and checks semantic relevance between claims and cited references.

SUGGESTED DATASETS / RESOURCES

OpenAlex; Crossref; Semantic Scholar; Zotero-compatible bibliographic data; publicly available research papers.

AI / ML COMPONENT

NLP; document classification; claim detection; semantic similarity; citation recommendation; information retrieval; LLM-based academic feedback.

WHAT MAKES THE CHALLENGE INTERESTING?

Directly connects AI with academic research and writing and supports writing quality and evidence-based citation practices.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload draft structure/clarity feedback claims flagged citation-needed suggestions cited-reference relevance source view.

TECHNICAL REQUIREMENTS

Python; document parsing; NLP; citation/reference parsing; semantic retrieval; evaluation; source traceability; user interface.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Analyze a research draft.
- Identify potentially unsupported claims/citation needs.
- Provide reference-relevance checks with traceable sources.

OPTIONAL ADVANCED FEATURES

- Add citation-style validation.
- Compare multiple reference candidates.
- Add section-level academic-structure analysis.
- Add claim-to-source evidence visualization.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use lawful/public or explicitly permitted research resources. Respect copyright, database terms and API limits. Clearly distinguish generated suggestions from verified research facts, preserve source attribution, and avoid fabricating citations, authors, findings or research gaps.

INNOVATION ANGLE

Treats citation support as evidence verification rather than simple reference generation.

SUCCESS / EVALUATION METRICS

Claim detection precision/recall, citation recommendation relevance, reference-support accuracy and user-facing feedback quality.



CHALLENGE 03

AI Research Topic & Problem Statement Generator

DOMAIN

Research Ideation / Research Design

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Students and new researchers often have broad ideas but struggle to convert them into focused, feasible, and researchable problems. They may also find it difficult to define research questions, objectives, scope, and possible novelty based on existing literature.

EXPECTED SOLUTION

Develop an AI tool that takes a user's research field, interests, skills, and available resources as input and generates focused research topics, problem statements, research questions, and objectives. The system should also retrieve related literature and highlight possible research gaps.

SUGGESTED DATASETS / RESOURCES

OpenAlex; Semantic Scholar; Crossref; arXiv for relevant fields; publicly available research abstracts/full-text datasets.

AI / ML COMPONENT

NLP; semantic retrieval; embeddings; topic clustering; LLM-based generation; similarity checking; basic research-gap analysis.

WHAT MAKES THE CHALLENGE INTERESTING?

Transforms a vague research idea into a structured, testable and research-ready concept.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter field/interests/resources generate topic candidates related literature problem statement research questions/objectives novelty/gap evidence.

TECHNICAL REQUIREMENTS

Python; retrieval; embeddings; LLM generation; similarity checking; structured output; source attribution; interface.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Generate focused topic/problem candidates.
- Generate research questions/objectives.
- Link suggestions to related literature and indicate possible gaps.

OPTIONAL ADVANCED FEATURES

- Add feasibility checks based on available resources.
- Compare generated ideas against existing literature.
- Add novelty/similarity warnings.
- Add user-controlled scope refinement.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS



Use lawful/public or explicitly permitted research resources. Respect copyright, database terms and API limits. Clearly distinguish generated suggestions from verified research facts, preserve source attribution, and avoid fabricating citations, authors, findings or research gaps. Generated research ideas must be treated as suggestions requiring human verification; the system should not claim novelty without evidence.

INNOVATION ANGLE

Links idea generation to evidence retrieval and feasibility rather than producing isolated text.

SUCCESS / EVALUATION METRICS

Relevance, novelty/similarity warning quality, source traceability, feasibility feedback and user evaluation.



CHALLENGE 04

AI Research Paper Summarizer & Key Findings Extractor

DOMAIN

Academic Summarization / Information Extraction

DIFFICULTY

Intermediate

PROBLEM STATEMENT

Researchers often need to read dozens of papers during literature review, but reading every paper in full can be extremely time-consuming. Important information such as research objectives, methodology, datasets, key findings, limitations, and conclusions may be difficult to extract consistently.

EXPECTED SOLUTION

Create an AI system that accepts research papers and automatically generates structured summaries containing the research objective, methodology, dataset, major findings, limitations, and conclusion. The system should allow users to compare summaries across multiple papers.

SUGGESTED DATASETS / RESOURCES

arXiv; PubMed Central; Semantic Scholar; OpenAlex; other publicly available research-paper datasets.

AI / ML COMPONENT

Document parsing; NLP; extractive and abstractive summarization; information extraction; embeddings; LLM-based structured output.

WHAT MAKES THE CHALLENGE INTERESTING?

Solves a common research problem and can reduce the time required to understand and compare large collections of papers.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload/select papers structured summary cards compare objectives/methods/datasets/findings/limitations source-linked evidence.

TECHNICAL REQUIREMENTS

Python; PDF/document parsing; NLP; summarization; structured extraction; embeddings; comparison interface; evaluation.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Generate structured summaries with the required fields.
- Support comparison across multiple papers.
- Keep summaries traceable to source sections where feasible.

OPTIONAL ADVANCED FEATURES

- Add cross-paper comparison tables.
- Add contradiction/consistency detection.
- Add section-level evidence links.
- Compare extractive and abstractive approaches.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use lawful/public or explicitly permitted research resources. Respect copyright, database terms and API limits. Clearly distinguish generated suggestions from verified research facts, preserve source attribution, and avoid fabricating citations, authors, findings or research gaps. Summaries should not replace reading the original paper for critical decisions; preserve source attribution and avoid invented findings.

INNOVATION ANGLE

Combines structured extraction and cross-paper comparison instead of a generic one-paragraph summary.

SUCCESS / EVALUATION METRICS

Summary factuality, coverage of required fields, faithfulness to source, retrieval/evidence quality and comparison usefulness.



CHALLENGE 05

AI Research Collaboration & Paper Matching Platform

DOMAIN

Research Collaboration /
Recommendation

DIFFICULTY

Intermediate to Advanced

PROBLEM STATEMENT

Students and researchers often have difficulty finding collaborators with complementary research interests and skills. Existing networks may contain information about people but do not always provide an effective way to match research interests, technical skills, and project needs.

EXPECTED SOLUTION

Develop an AI-based platform that matches researchers or students based on research interests, skills, previous projects, publications, and preferred research areas. The system should generate a compatibility explanation and suggest potential collaboration topics.

SUGGESTED DATASETS / RESOURCES

OpenAlex author/publication data; Semantic Scholar author profiles; publicly available researcher profiles; ORCID public data where permitted; synthetic profiles for hackathon testing.

AI / ML COMPONENT

Recommendation systems; NLP; text embeddings; semantic similarity; clustering; ranking algorithms; explainable matching.

WHAT MAKES THE CHALLENGE INTERESTING?

Combines research discovery with AI-based recommendation and can support interdisciplinary collaboration.

SUGGESTED ON-SCREEN DEMONSTRATION

Enter researcher profile ranked matches compatibility explanation overlapping interests/skills suggested collaboration topics.

TECHNICAL REQUIREMENTS

Python; profile/text processing; embeddings; ranking/recommendation; explainable matching; privacy-aware data handling; dashboard.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Generate researcher/collaborator matches.
- Provide a compatibility explanation.
- Suggest at least one potential collaboration topic per match.

OPTIONAL ADVANCED FEATURES

- Add publication-topic graphs.
- Add diversity-aware or interdisciplinary matching.
- Allow user feedback and re-ranking.
- Compare recommendation strategies.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use lawful/public or explicitly permitted research resources. Respect copyright, database terms and API limits. Clearly distinguish generated suggestions from verified research facts, preserve source attribution, and avoid fabricating citations, authors, findings or research gaps. Use only public/permitted researcher information and avoid inferring sensitive personal attributes. Provide clear controls for profile data and attribution.

INNOVATION ANGLE

Makes matching explainable and research-topic oriented rather than a generic similarity score.

SUCCESS / EVALUATION METRICS

Top-k recommendation relevance, match diversity/coverage, explanation quality and user feedback.